Human Resources Capstone Project

Project Title

Employee Performance Dashboard

Project Overview

The Employee Performance Dashboard is a data-driven HR analytics application designed to track, analyze, and visualize employee performance metrics within an organization. The system helps HR teams and managers monitor productivity, attendance, task completion, and overall employee performance through interactive dashboards and reports.

This project gives students practical exposure to HR analytics and workforce management systems used in modern organizations.

Project Objective

Develop an HR analytics application that can:

- Store employee performance data
- Analyze productivity and attendance metrics
- Track task and project performance
- Generate employee performance reports
- Visualize insights through dashboards

Industry Relevance

This project is inspired by systems used in:

- HR departments
- Corporate organizations
- Workforce management platforms
- Employee monitoring systems
- HR analytics tools

Students will understand how analytics is used in employee performance evaluation and decision-making.

Project Modules

Module 1: Employee Data Management

- Add employee records
- Store department and role details
- Manage attendance and task data
- Update employee information

Learning Outcomes

- Database management
- CRUD operations
- HR data handling

Module 2: Performance Tracking System

Track employee metrics such as:

- Attendance percentage
- Task completion rate
- Productivity score
- Project contribution
- Work efficiency

Learning Outcomes

- Data processing
- KPI tracking
- Performance evaluation logic

Module 3: HR Analytics Dashboard

Display:

- Employee performance scores
- Department-wise analysis
- Attendance trends
- Top performers
- Monthly performance reports

Learning Outcomes

- Dashboard development
- Data visualization
- Business reporting

Module 4: Performance Insights & Recommendations

Generate insights such as:

- Employees needing improvement
- High-performing employees
- Attendance warnings
- Productivity suggestions

Learning Outcomes

- HR analytics
- Insight generation
- Reporting systems

Module 5: Report Generation

Generate downloadable:

- Employee reports
- Monthly summaries
- Performance charts

Learning Outcomes

- Report automation
- Business documentation
- Data presentation

Recommended Tech Stack

Frontend

- HTML
- CSS
- Bootstrap or React.js

Backend

- Python Flask / Node.js

Database

- MySQL or MongoDB

Visualization

- Chart.js
- Power BI (Optional)

Deployment

- Render
- Netlify
- Vercel

Project Workflow

1. Employee Data Entry
2. Attendance & Performance Tracking
3. Data Processing
4. Dashboard Visualization
5. Performance Analysis
6. Report Generation

Dataset Suggestions

Students may use:

- Sample employee datasets
- HR analytics datasets from Kaggle
- Self-created employee performance data

Deliverables

Students must submit:

- Source Code
- Project Report
- PPT Presentation
- GitHub Repository
- Dashboard Screenshots
- Deployment Link
- Demo Video

Scope Limitations

Students should avoid:

- Real-time employee tracking systems
- Enterprise HRMS integrations
- Payroll automation
- Advanced AI-based monitoring systems

Focus should remain on building a clean and functional HR analytics dashboard.

25-Day Project Plan

Days	Tasks
Day 1–3	Requirement Analysis and UI Planning
Day 4–7	Employee Data Management Module
Day 8–12	Performance Tracking System
Day 13–17	Dashboard Development
Day 18–21	Insights and Report Generation
Day 22–23	Testing and Data Validation
Day 24	Bug Fixing and Optimization
Day 25	Deployment and Final Presentation

Evaluation Criteria

Criteria	Weightage
Functionality	30%
Dashboard & Analytics	25%
Database Management	15%
UI/UX Design	15%
Deployment	10%
Documentation	5%

Final Outcome

By completing this project, students will learn:

- HR analytics fundamentals
- Employee data management
- Dashboard development
- Data visualization techniques
- Report generation
- End-to-end web application workflow

The final system will function as an Employee Performance Dashboard capable of tracking workforce performance, generating insights, and visualizing employee analytics for HR decision-making.